Substitutes and Complements under Product Returns

Draft notes

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1 Model

A seller offers two goods separately and as a bundle. The price of good $i \in \{1, 2\}$ is p_i. The bundle price is p_B, with $\max(p_1, p_2) \leq p_B \leq p_1 + p_2$. Selling each good alone and selling the two together is mixed bundling. Producing a unit costs c. A returned unit costs h to handle.

A unit mass of consumers each value good i at v_i. The values are drawn independently from the uniform distribution on $[0, 1]$ and are known before buying. Good i works with probability α_i. Let W_i be the indicator that equals 1 when good i works and equals 0 otherwise, so $\alpha_i = \mathbb{E}[W_i]$. If good i works, the consumer receives v_i. If it does not work, the value is zero. The values are independent of (W_1, W_2).

Let $\sigma_i = \sqrt{\alpha_i(1 - \alpha_i)}$ be the standard deviation of W_i, and let ρ be the correlation of W_1 and W_2. The probability that both goods work is $\alpha_1\alpha_2 + \rho\sigma_1\sigma_2$. The probability that only good 1 works is $\alpha_1(1 - \alpha_2) - \rho\sigma_1\sigma_2$. The probability that only good 2 works is $\alpha_2(1 - \alpha_1) - \rho\sigma_1\sigma_2$. The probability that neither works is $(1 - \alpha_1)(1 - \alpha_2) + \rho\sigma_1\sigma_2$. These four probabilities are positive. When $\rho = 0$ the goods work independently. A positive ρ means the goods are complements. They tend to work together. A negative ρ means the goods are substitutes. One tends to work when the other does not.

The consumer buys good 1, good 2, the bundle, or nothing, before learning (W_1, W_2). A consumer who keeps only good i pays p_i. A consumer who keeps both pays p_B. A consumer who keeps nothing pays nothing. The payoff from buying good i alone is $\max(W_i v_i - p_i,\, 0)$. The payoff from buying the bundle is

$$\max\big(W_1 v_1 + W_2 v_2 - p_B,\; W_1 v_1 - p_1,\; W_2 v_2 - p_2,\; 0\big).$$

The consumer chooses the option with the highest expected payoff, and buys nothing when that payoff is negative.

Buying the bundle and then keeping one good is bracketing. It is more attractive when ρ is low, because one good then works when the other does not.

The seller pays c for each good the consumer buys and h for each good the consumer returns, and collects the payment above. Profit is the expected value of revenue minus these costs.